

# Farmhannong Agro Weekly - 출처 확인용 PDF

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| 카드 발행일    | 2026 - 08 - 24                                                                                                                                                                                                                                                                                        |
| 카드 번호     | 16                                                                                                                                                                                                                                                                                                    |
| 출처        | ScienceDaily (UC Davis, Current Biology)                                                                                                                                                                                                                                                              |
| 원문 제목     | Queen bees have a disturbing way to survive pesticides                                                                                                                                                                                                                                                |
| 원 URL     | <a href="https://www.sciencedaily.com/releases/2026/08/260821012224.htm">https://www.sciencedaily.com/releases/2026/08/260821012224.htm</a>                                                                                                                                                           |
| 제목 검색 URL | <a href="https://www.google.com/search?q=%22Queen+bees+have+a+disturbing+way+to+survive+pesticides%22+ScienceDaily+%28UC+Davis%2C+Current+Biology%29">https://www.google.com/search?q=%22Queen+bees+have+a+disturbing+way+to+survive+pesticides%22+ScienceDaily+%28UC+Davis%2C+Current+Biology%29</a> |
| 검색 키워드    | queen bee maternal offloading pesticide eggs methyl parathion UC Davis Current Biology 2026 Nicklisch worker bees filter 95% colony collapse August 21 2026                                                                                                                                           |
| PDF 생성일   | 2026 - 08 - 24 08:54                                                                                                                                                                                                                                                                                  |

## 카드 요약 / 당시 확인 내용

Researchers at the University of California, Davis, working with Lawrence Livermore National Laboratory and the USDA Agricultural Research Service, found that queen bees exposed to pesticides transfer contamination from their bodies into their eggs, a process described as maternal offloading. The study, published in Current Biology 2026;36(15):3734, used methyl parathion tagged with radioactive markers at environmentally relevant but non-lethal concentrations. Worker bees filtered 95% of pesticides on day one, declining to 86% by day 10. Researcher Sascha Nicklisch said that in order to protect herself, the queen bee offloads these chemicals into her eggs to get rid of them. Because queens produce 1,500 - 2,000 eggs daily, accumulating contamination in eggs could reach a tipping point where developing larvae fail to develop properly, potentially contributing to delayed colony collapse.